Isolation of early Paleohelikian remanence in Grenville anorthosites of the French River area, Ontario

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Specimens were collected at 19 sites from Precambrian anorthosites in the French River region of the Grenville Province. AF and thermal step demagnetization analyses indicate the presence of four components: A, B, C, and D. After bulk AF and thermal demagnetization at 550, 650, and 670°C, conventional two-tiered statistical analysis reveals only the presence of the B component for the collection. Point density contour plots isolate objectively all four components. Vector subtraction shows that the A component survives up to ~40 ± 10 mT or 300 ± 100°C and resides in magnetite. Its pole position of 45°W, 26°N, δp = 9°, δm = 10° is consistent with it being a Grenvillian orogenic metamorphic overprint acquired 975 Ma ago at the same time as the K–Ar dates were reset. The B component survives up to 100 mT and resides both in magnetite up to 585°C and in hematite up to 670°C. Its pole position of 26°W, 13°S, δp = 1°, δm = 2° is ~30° off the pre-1000 Ma portions of the APW path, but falls directly on the 1725 Ma portion. This fits the known whole-rock Rb–Sr and zircon U–Pb ages of 1700 ± 100 Ma found for these rocks, which date the Hudsonian Orogeny. The C component is found only at 650°C or above. Its pole of 57°W, 27°S, δp = 4°, δm = 7° is 60° away from 1000–1400 Ma portion of the APW path but falls directly on the 1800 Ma path. The D component is isolated only after the 670°C treatment. Its pole of 1°W, 6°N, δp = 4°, δm = 7° falls on the 1250 Ma segment of the APW path. It is attributed to alteration associated with the Sudbury olivine diabase dikes of this age, which are adjacent to some sites. Thus the Grenville Orogeny is essentially only a thermal event in the anorthosites, and the B and C components were acquired during cooling after the more intense Hudsonian Orogeny. Also population level screening provides a more efficient and objective way of isolating multicomponents than conventional tiered statistics after bulk AF or thermal cleaning of a collection.

Des spécimens d'anorthosites précambriennes ont été prélevés à 19 sites dans la région de French River de la Province du Grenville. Des analyses de désaimantation par étapes utilisant le traitement par champ alternatif et le traitement thermique indiquent l'existence de quatre composantes: A, B, C et D. Après la désaimantation globale par champ alternatif et chauffage à 550, 650 et 670°C, une analyse statistique générale à deux facteurs conventionnelle révèle seulement l'existence de la composante B pour l'ensemble des spécimens. Des courbes d'égales densités de points mettent en évidence de manière objective les quatre composantes. La soustraction du vecteur montre que la composante A persiste jusqu'à ~40 ± 10 mT ou 300 ± 100°C et elle est portée par la magnétite. La position de son pôle à 45°O, 26°N, δp = 9°, δm = 10° correspond à une surimpression d'origine métamorphique lors de l'orogénèse du Grenvillien il y a 975 Ma coïncidant avec la remise à zéro pour les datations au K–Ar. La composante B persiste jusqu'à 100 mT et est portée à la fois par la magnétite jusqu'à 585°C et par l'hématite jusqu'à 670°C. La position de son pôle à 26°O, 13°S, δp = 1°, δm = 2° est éloignée d'environ 30°C des segments inférieurs à 1000 Ma du tracé de la dérive apparente des pôles, mais tombe directement sur le segment 1725 Ma. Ces données concordent avec les âges mesurés sur les roches totales par le Rb–Sr et l'U–Pb du zircon à 1700 ± 100 Ma, ce sont ces roches qui datent l'orogénèse de l'Hudsonien. La composante C est observée uniquement à 650°C ou plus. Son pôle à 57°O, 27°S, δp = 4°, δm = 7° est 60° éloigné du segment 1000–1400 Ma du tracé de la dérive apparente des pôles mais tombe directement sur la courbe 1800 Ma. La composante D apparaît seulement après un traitement thermique à 670°C. Son pôle à 1°O, 6°N, δp = 4°, δm = 7°, se situe sur le segment 1250 Ma du tracé de la dérive apparente des pôles. Elle est attribuée à une altération relative aux dykes de diabase à olivine de Sudbury de cet âge qui se retrouvent adjacents à quelques sites. Ainsi l'orogénèse du Grenvillien résulte essentiellement d'un phénomène de réchauffement dans les anorthosites, et les composantes B et C ont été introduites lors du refroidissement qui a suivi l'orogénèse de l'Hudsonien plus intense. Aussi la sélection de niveaux de population constitue une méthode plus efficace et plus objective pour mettre en évidence les composantes multiples que la méthode habituelle d'analyse statistique générale de facteurs après les traitements de nettoyage global par champ alternatif ou par chauffage d'un groupe de spécimens.

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Introduction

Radiometric evidence from rocks in the French River area of the Precambrian Grenville Province indicates that they have undergone at least two major orogenic episodes of metamorphism and deformation. The most recent Grenville Orogeny reset the K–Ar decay systems about 950 Ma ago (Krogh and Davis 1968; Harper 1967; Wynne-Edwards 1972). Heretofore, the paleomagnetic ages of the Grenville rocks have been interpreted to be metamorphic overprints that were reset at the same time as the K–Ar ages (Buchan and Dunlop 1976; McWilliams and Dunlop 1978). The resetting of the K–Ar

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system by argon loss may occur on reheating at temperatures as low as ~200°C over time intervals of the order of ~50 Ma (Dalrymple and Lanphere 1969; Moorbath 1967). Pullaiah *et al.* (1975) have shown that magnetite and hematite with blocking temperatures of ~410 and ~520°C, respectively, would be reset under the same conditions. However, the most stable magnetite and hematite remanence carriers have blocking temperatures approaching 585 and 675°C, respectively, that are not reset unless these temperatures are actually reached. Thus, the remanence in these most stable carriers may well predate the Grenville Orogeny.

The Hudsonian Orogeny is the earlier major episode affecting the study area. It is indicated by whole-rock Rb–Sr and zircon U–Pb dates of 1700 ± 100 Ma (Krogh and Davis 1968). In this work the Grenville episode is detected by limited isotopic exchange between adjacent minerals.

The sampled anorthosite plutons occur either closely associated with monzonitic to granitic rocks or as small lenses in migmatitic gneisses (Ontario Department of Mines 1974). These gneissic, interlayered (rarely massive) anorthosites have the same gneissosity as the associated monzonitic to granitic rocks and the migmatitic gneisses, which suggests a common deformation and metamorphic history.

An additional complexity in analyzing the remanence components is the possibility that metamorphic chemical alteration during any episode may well produce hematite with a highly stable chemical remanent magnetization (CRM) component.

Statistical analysis of remanence data

The introductory discussion illustrates the fact that the isolation and identification of multiple remanence components in metamorphic rocks are difficult problems. In this study, three methods of statistical analysis are used and compared. These are conventional two-tiered screening, specimen level screening, and population level screening.

Conventional two-tiered statistical analysis uses the two specimen directions as a measure of the core remanence homogeneity. Accepted homogeneous core mean directions are then used as a measure of the site remanence homogeneity. Accepted homogeneous site means are then used to compute the component mean direction.

Specimen level screening uses the 12 magnetic measurements generated in the six-spin magnetometer measurement mode. It provides $n = 4$ measurements for each of the three Cartesian remanence components (x, y, z) of the specimen, which are then used to calculate the specimen's angular standard deviation (ASD) following the general method of Reed (1959). The best estimate of the variance, S_r^2, of the specimen direction $\boldsymbol{r} = (\bar{x}\boldsymbol{N} + \bar{y}\boldsymbol{E} + \bar{z}\boldsymbol{V})/R$ is

$$S_r^2 = (57.3)^2\left[\left(\frac{\partial \boldsymbol{r}}{\partial x}\right)^2_{\bar{x},\bar{y},\bar{z}} S_x^2 + \left(\frac{\partial \boldsymbol{r}}{\partial y}\right)^2_{\bar{x},\bar{y},\bar{z}} S_y^2 + \left(\frac{\partial \boldsymbol{r}}{\partial z}\right)^2_{\bar{x},\bar{y},\bar{z}} S_z^2\right](^\circ)^2$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$S_x^2 = [1/(n-1)]\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$R = (\bar{x}^2 + \bar{y}^2 + \bar{z}^2)^{1/2}$$

and $\boldsymbol{N}$, $\boldsymbol{E}$, and $\boldsymbol{V}$ are unit vectors along the north, east, and vertical directions, respectively. Explicitly, the $\mathrm{ASD} = S_r = (57.3/R^3)[(R^2 - \bar{x}^2)^2 S_x^2 + (R^2 - \bar{y}^2)^2 S_y^2 + (R^2 - \bar{z}^2)^2 S_z^2]^{1/2}$ and the radius of the circle of 95% confidence about $\boldsymbol{r}$ is $\theta_{95} = 1.96\mathrm{ASD}/(n)^{1/2} \approx \mathrm{ASD}$ for $n = 4$. This calculation is most valid when S_x, S_y, and S_z are small relative to $\bar{x}$, $\bar{y}$, and $\bar{z}$ as are those of Harrison (1980) and Briden and Arthur (1981). Harrison (1980) used the same mathematical approach; however, he calculated the ASD for the declination and inclination independently. The single parameter estimate of precision of the measured remanence defined above gives information equivalent to his two parameter estimate. The maximum limiting value (Harrison 1980) for acceptance of a specimen direction is an ASD ≈ 35° above which there is an estimated 5% probability that the specimen retains no coherent magnetization. In practice a much lower value of ASD (<20°) is preferable depending on the subsequent planned mathematical treatment for the data. Thus if the proposed treatment includes least-squares component fitting, vector subtraction, or remagnetization circle fitting (Stupavsky and Symons 1978; Fuller 1969; Halls 1975) an ASD <5° may be appropriate to avoid large error limits. Finally, for specimen level screening, all homogeneously magnetized specimens that do not exceed the ASD screening criteria are used to calculate the site mean remanence direction.

Population level screening uses all specimen directions after a common demagnetization treatment. The directions are plotted on both hemispheres of a stereonet and then the point density is contoured using a conventional 1% area smoothing circle and percentage contour intervals. Concentrations of directions called anomalies identify the various remanence components. The screening is achieved by selecting only those specimen directions that explicitly define a remanence component (i.e., are enclosed within a given contour defining the

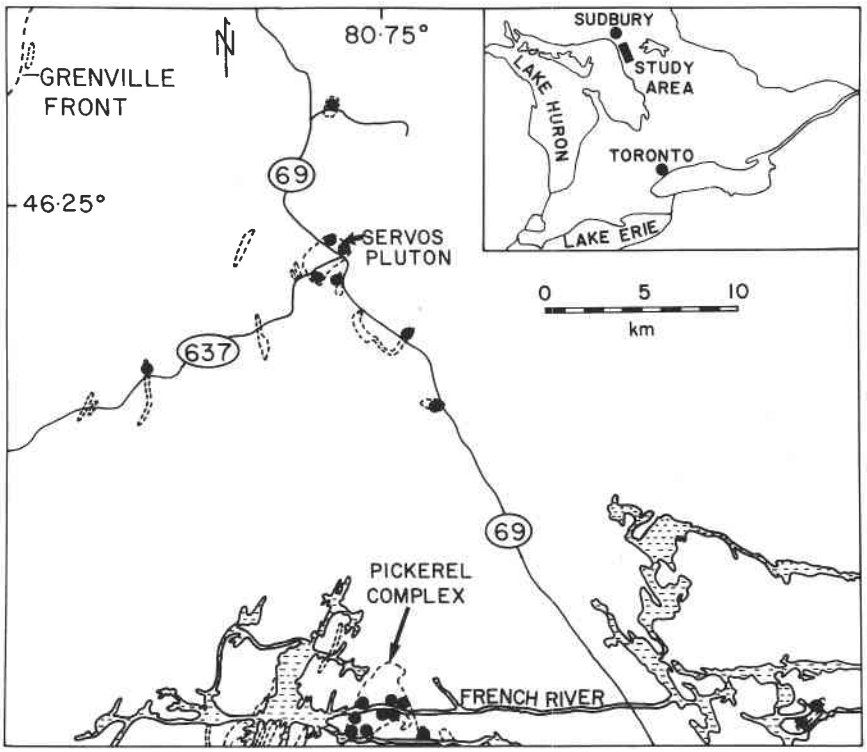


FIG. 1. Location of the study area and the sampled sites. Dashed lines outline the anorthosite bodies in the study area.

anomaly) for the computation of its mean and precision. The statistical significance of any anomaly may be estimated directly from the contour plot when Kamb's (1959) method of smoothing and contouring is used. This is similar to the "mode" computed by Van Alstine (1980) for use in analyzing paleomagnetic vectors. He concludes that the mode of a streaked or girdled distribution of vectors is not affected by outlier vectors, and therefore it is more representative of the characteristic magnetization than the conventional Fisherian mean.

Sampling and measurement

The Grenville anorthosites were sampled at 19 sites (Fig. 1) by drilling five cores at each site. Two or more 2.4 cm long specimens were cut from each 2.5 cm diameter core. The natural remanent magnetization (NRM) of each core was measured using a Schonstedt SSM-IA magnetometer. One pilot specimen per site was alternating field (AF) step demagnetized in 10 steps up to 100 mT using a Schonstedt GSD-1 demagnetizer. A further nine specimens, making two per core, from each site were AF demagnetized at a specific intensity between 20 and 100 mT based on the behaviour of the pilot specimen. A new selection of 16 fresh pilot samples from 16 sites was made for thermal step demagnetization in 10 steps up to 670°C using a non-inductive shielded furnace. Finally all the specimens from the 19 sites that had homogeneous remanence remaining after AF demagnetization were thermally demagnetized further at 550, 650, and 670°C.

Step demagnetization

Specimen level screening was used in analyzing the data from the pilot specimens on step demagnetization. Of the 19 AF and 16 thermal pilot specimens, 18 and 13 were accepted with an ASD $<6°$ on treatment up to 100 mT or 550°C, respectively. For the thermal specimens, the selection criterion was relaxed to ASD $<20°$ for the 650 and 670°C steps. In addition, two of the accepted AF pilot specimens had extreme outlier directions and were rejected to leave 16 accepted specimens.

The magnetic carriers in the accepted specimens have a broad coercivity range of NRM (Fig. 2). On average 60% of the NRM resides in carriers having coercive release forces of <10 mT and 30% of >100 mT. The blocking temperatures show a broad range also with 55% of the NRM removed by ~300°C and the final 40% removed between 500 and 675°C. Four of the pilot specimens were particularly stable with >80% of their NRM remaining after 100 mT or 550°C.

The mean remanence directions of the pilot specimens on AF and thermal step demagnetization indicate

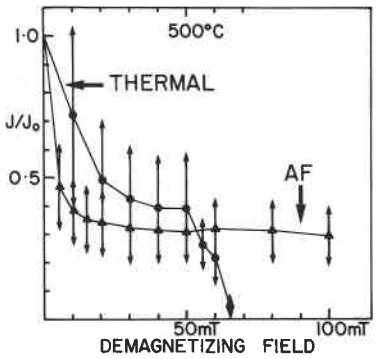


FIG. 2. Mean remanence intensity of pilot specimens from different sites during AF and thermal step demagnetizations. The mean NRM intensities are: 2.91×10^{-1} A m^{-1} cm^{-3} for 16 AF and 2.87×10^{-1} A m^{-1} cm^{-3} for 15 thermal pilot specimens. The vertical bars indicate the standard deviation limits.

the presence of two well defined remanence components (Figs. 3, 4; Table 1). The mean vector removed between the selected demagnetizing steps shows that a magnetically softer A component is being removed initially up to at least 40 mT and 300°C after initial removal of the viscous remanent magnetization (VRM) component (Fig. 4*a*). The mean A components being removed between the six selected demagnetizing steps have relatively high α_{95} values of ~30° (Fisher 1953) and are statistically identical. Their mean at 112°, 71°, 6° (declination, inclination, α_{95}; Table 1) is likely the most reliable estimate of the direction of the A component determined in this study.

The B component is indicated by a moderately stable end point between 50 and 100 mT on AF demagnetization (Figs. 3 and 4*b*). At these AF steps the individual specimen vectors removed between successive demagnetizing steps form a highly dispersed population because of the acquisition of small spurious magnetization components by many of the specimens during the AF demagnetization process. The B component at 127°, 31°, 5° is better isolated by thermal demagnetization between 500 and 600°C. The mean vector removed (Table 1) shows its removal in three thermal steps between 500, 550, 600, and 650°C.

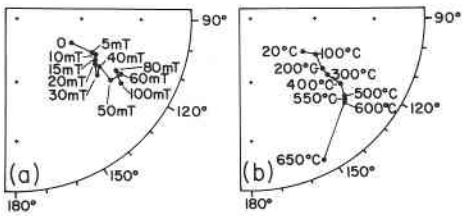


FIG. 3. Mean remanence directions, after each demagnetizing field, of pilot specimens are shown on a Schmidt equal-area stereoplot with solid symbols denoting down directions. The radii of circles of 95% confidence for selected AF steps are: 19.1° at NRM, 13.1° at 5 mT, 11.7° at 10 mT, and 13.1° at 100 mT; for thermal steps are: 13.1° at 20°C, 15.1° at 200°C, 10.5° at 500°C, 12.5° at 550°C, 11.8° at 600°C, and 18.4° at 650°C.

The thermal step demagnetization data at 650°C indicate the existence of a third C component at 154°, 8°, 18° residing dominantly in hematite carriers having blocking temperatures in excess of 650°C. At the next thermal step of 670°C only a few of the pilot specimens remained magnetically homogeneous and these gave the B direction. The directions of the inhomogeneous specimens were highly dispersed and formed no meaningful mean direction.

Conventional core–site screening

Conventional two-tiered statistical analysis of the paleomagnetic data requires remanence direction homogeneity at the core and site levels for acceptance. It isolates only the B component. After AF demagnetization, the B component is isolated in 14 sites giving an α_{95} of 7° (Table 2). After thermal demagnetization at 550, 650, and 670°C, the B component is isolated in 11, 10, and two sites, respectively, with α_{95} values of 5, 8, and 36°. Neither of the A or C components is isolated.

Specimen level screening

When specimen level screening is used, followed by site level screening, the B component is isolated at 14 sites with an α_{95} of 7° on AF demagnetization (Table 2). Thus there is no improvement compared with the conventional tiered mode of analysis. After thermal demagnetization, however, there is a marked improvement in accepted sites and confidence giving isolation of B at 550, 650, and 670°C in 14, 11, and five sites with α_{95} of 4, 6, and 10°, respectively. But, as before, neither of the A nor C components is isolated.

Population level screening

When the NRM directions of all 199 measured specimens are plotted on a stereonet, smoothed, and contoured (Fig. 5*a*), an elongated anomaly is produced that contains two peaks linked by an intervening girdle. This shows that the mode of the NRM population is composed of a dominant A component approximately defined by its 15% density contour at 134°, 75°, and a B component defined by its 7.5% contour at 126°, 40°. The girdle is defined by hybrid components of A and B. More reliable estimates of the mean A and B directions are obtained (Table 2) through population level screening. The 93 directions defining the A component and the 35 directions defining the B component, which are enclosed within the nearly circular 7.5% contour, are used to compute each mean and its precision using Fisherian statistics and giving each direction unit weight. These mean A and B directions agree with those

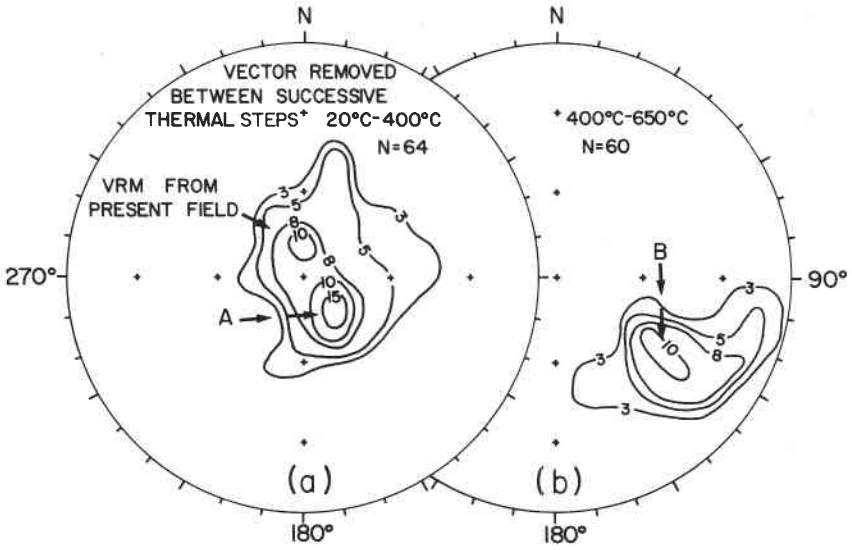


FIG. 4. Point density stereocontour plots of the vector removed between successive thermal demagnetization steps of all homogeneously magnetized specimen directions from: (*a*) 20 to 400°C; and (*b*) 400 to 650°C. A conventional 1% smoothing circle was used and the contour intervals are in percents; *N* is the number of vector removed directions plotted.

TABLE 1. Mean vector removed between selected demagnetizing steps for which $\alpha_{95} < 35°$

Demagnetizing step	No. of pilot specimens [steps]	Mean vector removed decl. (°)	incl. (°)	α_{95} (°)	Removed component
Alternating field					
(1) NRM—5 mT	16	116	72	32	A
(2) 5–10 mT	16	116	60	25	A
(3) 20–40 mT	16	102	74	26	A
Thermal					
(4) NRM—100°C	15	134	70	28	A
(5) 100–200°C	15	90	73	30	A
(6) 200–300°C	15	109	73	30	A
(7) 500–550°C	13	129	32	23	B
(8) 550–600°C	11	124	29	25	B
(9) 600–650°C	11	129	33	30	B
Mean removed					
A component	[6]	112	71	6	A
B component	[3]	127	31	5	B

isolated by step demagnetization (Table 2). The components are better defined if specimen level screening is applied prior to population level screening and therefore this combination is used subsequently. It should be noted that the mean direction of a magnetization component determined by population level screening or filtering on the basis of contours is identical to the mode of the anomaly because all specimen directions enclosed by the lowest nearly circular contour are used. Thus this specimen selection method is the graphical equivalent to the elegant analytical method of Van Alstine (1980) for determining the precision of the mode. In his method, the precision of a mode is given by the Fisherian α_{95} of the mean of the largest subset of the distribution whose mean is identical to the mode.

After AF demagnetization at peak fields from 20 to

TABLE 2. Screened remanence data

Component	Treatment	Screening	No. of sites (specimens) [steps]	Decl. (°)	Incl. (°)	α_{95} (°)
A	AFs	VR	[3]	112	69	12
	Ts	VR	[3]	112	73	11
	NRM	C	8	110	66	14
	NRM	P	(93)	103	77	3
B	NRM	P	(35)	122	38	3
	AFs	EP	(16)	122	30	13
	Ts	VR	[3]	127	31	5
	AF	C	14	125	34	7
	AF	S	14	124	31	7
	AF	P	(63)	125	27	2
	T = 550°C	C	11	126	25	5
	T = 550°C	S	14	128	24	4
	T = 550°C	P	(90)	126	25	2
	T = 650°C	C	10	129	26	8
	T = 650°C	S	11	129	24	6
	T = 650°C	P	(57)	126	24	3
	T = 670°C	C	2	123	20	36
	T = 670°C	S	5	127	19	10
	T = 670°C	P	(28)	125	19	5
C	Ts	EP	(12)	154	8	18
	T = 650°C	P	(16)	158	26	7
	T = 670°C	P	(8)	157	25	6
D	T = 670°C	P	(6)	93	22	7

NOTES: C = core level screening; accepted cores have $R \geq 1.90$, accepted sites have $\alpha_{95} \leq 20°$ for AF, T = 550 and 650°C and $\alpha_{95} \leq 30°$ for T = 670°C. S = specimen level screening; accepted specimens have within specimen remanence angular standard deviation (ASD) $\leq 6°$ for a six-spin measurement for AF, T = 550°C and ASD $\leq 20°$ for T = 650 and 670°C, accepted sites have $\alpha_{95} \leq 20°$ for AF, T = 550 and 650°C and $\alpha_{95} \leq 30°$ for T = 670°C. P = population level screening; accepted specimen directions with ASD $\leq 6°$ for NRM, AF, T = 550°C and ASD $\leq 20°$ for T = 650 and 670°C are plotted on a stereonet, smoothed, and contoured; remanence components are defined by anomalies; population screening is achieved by selecting only those directions defining the anomaly to compute its mean direction and precision. VR = vector removed between selected alternating field step (AFs) and thermal step (Ts) demagnetization. EP = mean direction of pilot specimen at the end point of step demagnetization.

100 mT, 185 specimens proved magnetically homogeneous. These are plotted in Fig. 5*b*. The shift in the concentration of directions is clearly evident. The A component is severely diminished by its preferential removal and is no longer defined by an encircling contour. Some remaining A or A–B hybrid components are indicated by the tail on the anomaly defined by the 2.5 and 5% contours. The B component is enhanced significantly and is now defined by the 17.5% contour. The mean B direction computed by using 63 specimen directions enclosed within the 7.5% contour now aligns closely with the mean directions isolated by step demagnetization and conventional analysis (Table 2).

Thermal demagnetization at 550°C shows virtually total removal of the A component in the 163 accepted specimens (Fig. 5*c*). The B component is further enhanced and is now defined by the 25% contour. At 650°C (Fig. 5*d*) with 151 acceptable specimens, the B component is still well defined by the 15% contour. The C component that was indicated by the mean direction at the 650°C end point of thermal step demagnetization is defined by a 5% contour. Finally, at 670°C with 130 acceptable specimens, the B component remains well defined by the 7.5% contour. The 2.5% contour shows the presence of the C and D components as well as hybrid B–C and A–B components.

Statistical significance of the remanence components isolated from contour plots

The statistical significance of the remanence components isolated from a stereocontour plot of N acceptable remanence directions may be estimated by a method proposed by Kamb (1959). The method is based on determining the probability that a given anomaly may be

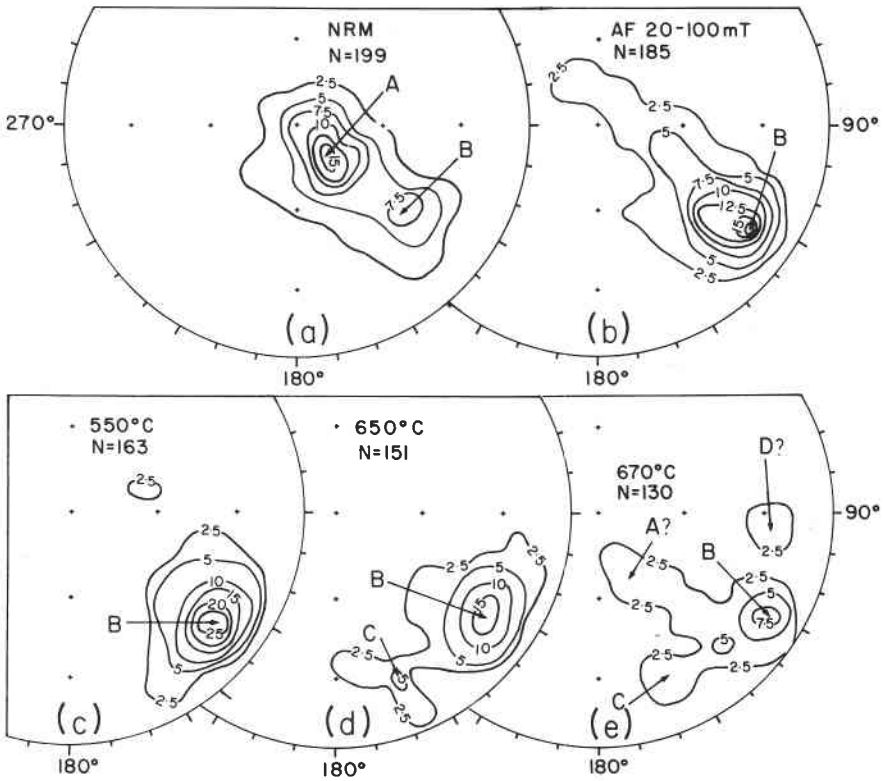


FIG. 5. Point density stereocontour plots of all homogeneously magnetized specimen directions (*a*) after NRM; (*b*) after AF demagnetization at optimum field; (*c*) after thermal demagnetization at 550°C; (*d*) after thermal demagnetization at 650°C; and (*e*) after thermal demagnetization at 670°C. *N* is the number of specimen directions plotted.

produced from N randomly directed vectors. When a smoothing circle of area A equal to 1% of the area of the stereonet is used to smooth a plot of N randomly directed vectors, then $E = NA$ directions are expected in each circle with a standard deviation of $\sigma = [NA(1 - A)]^{1/2}$, which simplifies to $\sigma = 0.1N^{1/2}$ when $A = 0.01$ (after Kamb 1959). Thus any anomaly is statistically significant at the 95% confidence level when defined by a closed contour of at least $E + 2\sigma$ or $0.01N + 0.20N^{1/2}$ directions and at the 99% confidence level when defined by a closed contour of at least $E + 3\sigma$ or $0.01N + 0.30N^{1/2}$ directions. This significance test for the least defined anomalies *C* and *D* in Fig. 4*e* gives a value of 3.6 directions for $N = 131$, which is equivalent to the 2.8% contour for 95% confidence. Since the anomaly peak values exceed a 2.8% density, both are significant at 95% confidence. It should be noted, however, that an anomaly may be statistically but not paleomagnetically significant. A reasonable additional requirement for paleomagnetic significance is that the anomaly must represent several sites with more than 5% of the population. This is true for the A, B, C, and D anomalies in this study.

Relative merits of population level screening for paleomagnetic remanence data

The stereocontour analysis presents graphically the directional distribution of the whole population and effectively shows the success of the various demagnetization treatments on the whole population. Because all the accepted reliable remanence data are plotted, any coherent magnetization isolated in only a few specimens shows up as an excess number of directions on the plot, which defines an anomaly that stands out relative to the background distribution or "noise." In this study, population level screening has identified and isolated successfully the well defined A and B components that have also been isolated by conventional analysis. In addition, after 650 and 670°C demagnetization, it has also isolated a third C magnetization that is also indicated by only the 650°C end point of thermal step demagnetization for some specimens and a fourth D component at 670°C. Thus, relative to conventional analysis, population level screening appears to have a significantly better ability to identify stable remanence components carried by only occasional specimens per site. These specimens are generally either rejected

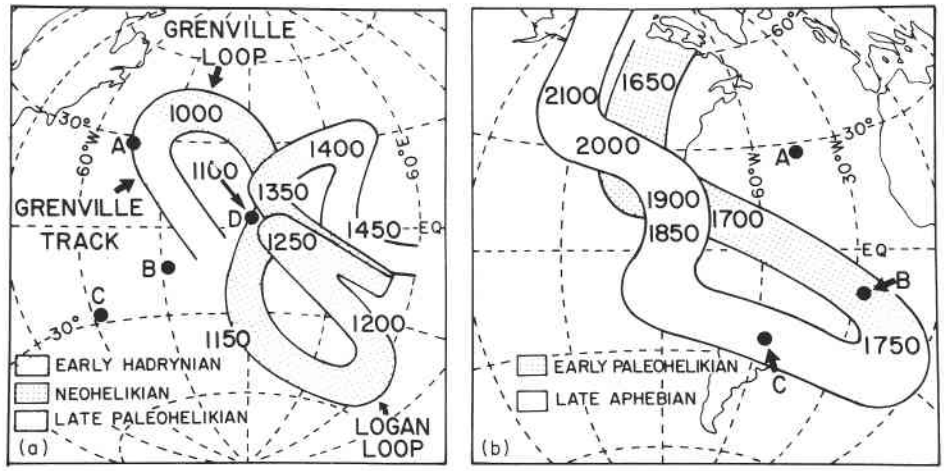


FIG. 6. VGP for the A, B, C, and D remanence components plotted on Irving's (1979) APW paths for (*a*) 950–1450 Ma; and (*b*) 1650–2200 Ma. The VGP for the D component is off the map of the 1650–2200 Ma path and is not shown.

TABLE 3. Virtual geomagnetic pole for the most reliable determination of A, B, C, and D remanence components from Tables 1 and 2

Component	Screening method	Virtual geomagnetic pole Long. (°W)	Lat. (°N)	δp (°)	δm (°)	Paleomagnetic age (Ma)
A	Vector removed, Table 1	45	26	9	10	975
B	Thermal, 550°C, P, Table 2	26	−13	1	2	1725
C	Thermal, 650°C, P, Table 2	57	−27	4	7	1800
D	Thermal, 670°C, P, Table 2	1	6	4	7	1250

during the screening process or their significance is lost when averaged with other accepted specimens to give a hybrid component direction.

Age of the remanence components

The most reliable estimate of the direction of the A component is probably the mean of the mean vectors removed during step demagnetization at 112°, 71°, 6°. Its pole position at 45°W, 26°N, δp = 9°, δm = 10° falls directly on the 975 Ma position of the apparent polar wander (APW) path of Irving (1979) for North America (Fig. 6*a*). The A pole does not fall on any segment of the APW path during the preceding 2500–1000 Ma time interval. Therefore the A component is interpreted to be a metamorphic remanence overprint formed during the Grenville Orogeny. Step demagnetization shows that this low AF and thermal coercivity component resides primarily in relatively coarse-grained magnetite or titanomagnetite. It was acquired at the same time as the K–Ar ages were reset. The poorly defined residual A and hybrid A–B components that survive 670°C demagnetization may be attributed to minor amounts of hematite alteration produced during the metamorphism.

The dominant B component is well defined by AF step, thermal step, and bulk AF and thermal demagnetizations. It is dominant in the most stable magnetite and hematite carriers up to 100 mT and to 585 and 670°C. It is best defined by thermal demagnetization at 550°C (Tables 2, 3). This direction gives a pole position located at 26°W, 13°S, δp = 1°, δm = 2°. This position is 30° of arc away from any segment of the Grenville APW path that predates 1000 Ma (Fig. 6*a*) and therefore was not produced by the Grenville Orogeny. The pole does, however, lie on the 1725 Ma segment of the APW path (Fig. 6*b*). This position agrees well with the 1700 ± 100 Ma Rb–Sr and U–Pb ages produced in the earlier more intense Hudsonian Orogeny affecting the study area. Therefore, the B component is thought to have been set during cooling of the anorthosites as the Hudsonian Orogeny waned.

The C component direction is best defined by thermal demagnetization at 650°C using population level screening. Its pole at 57°W, 27°S, δp = 4°, δm = 7° is 60° of arc from the pre-1000 Ma APW path of the Grenville Track. It does, however, lie directly on the 1800 Ma point of APW path (Fig. 6*b*). Existence of this compo-

nent could reflect a variety of possibilities. The most likely possibility is that it was formed slightly earlier during the Hudsonian Orogeny by either chemical alteration or higher temperature blocking during the cooling stage.

The D component, which is isolated after 670°C cleaning, gives a pole of 1°W, 6°N, $\delta p = 4°$, $\delta m = 7°$, which falls on the 1250 Ma segment of the APW path. Although not well defined, it is attributed to the presence of CRM carried by minor hematite formed by alteration during the intrusion of 1250 Ma Sudbury olivine diabase dikes. These dikes are known to occupy fracture zones forming the French River channel system along which the several sites containing D component directions were collected.

The generation of the B and C components in these Grenville Province anorthosites during the Hudsonian Orogeny is further strong evidence against the hypothesis of Irving *et al.* (1972) and Palmer and Carmichael (1972, 1973) that the Grenville Province was once a displaced terrain relative to the older Precambrian Shield.

Conclusions

Two significant conclusions result from this study.

First, point density stereocontour plotting of all reliable specimen remanence directions outperforms conventional two-tiered statistical analysis for isolating multicomponents. Collection or population level screening utilizes all the data and is efficient at objectively identifying the components as point density anomalies defined by a statistically significant contour interval. The mean direction and its statistical precision are readily computed by combining only those directional points that lie within the contour boundary. The method is particularly successful in identifying those components that reside in only a few specimens per site because, in conventional analysis, these component directions are generally rejected at the core or site level of screening.

The second significant result is that remanence components attributable to the Hudsonian Orogeny have been identified in these highly metamorphosed "Grenville" anorthosites of the French River area. The 1725 and 1800 Ma ages of the B and C components determined from the APW path agree well with the 1700 ± 100 Ma Rb–Sr and U–Pb zircon ages found in this area. This is the first study that has defined components of this age in Grenville Province rocks. Their Hudsonian age also provides strong evidence that the Grenville orogen was juxtaposed against the Southern and Superior orogens at this time. This conflicts with the hypothesis that the Grenville orogen was once a microplate that was rafted into place from a great distance by plate tectonics since the Hudsonian.

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